

CSC6033 - Week 6 Coding Project (P#6)

Turing Machine for Unary Addition

Irfan Ahmed

Unary Number Representation

In unary notation:

- A number n is represented by n occurrences of the symbol 0.
- Two numbers are separated by a special symbol, e.g., #.

Examples:

- $2 \rightarrow 00$
- $3 \rightarrow 000$
- $2 + 3 \rightarrow 00\#000 \Rightarrow \text{Result: } 00000$

Objective

Design a Turing Machine (TM) that:

- Accepts two unary numbers separated by #
- Replaces # with 0 to merge the two sequences
- Replaces the last 0 with a blank (B) to mark the end
- Halts after completing the addition

Formal Definition of the Turing Machine

$M = (Q, \Sigma, \Gamma, \delta, q_0, B, F)$

- $Q = \{q_0, q_1, q_2, q_3\}$ (Set of states)
- $\Sigma = \{0, \#\}$ (Input alphabet)
- $\Gamma = \{0, \#, B\}$ (Tape alphabet)

- δ = Transition function
- q_0 = Start state
- B = Blank symbol
- $F = \{q_3\}$ (Final state)

Transition Table

Current State	Read Symbol	Write Symbol	Move	Next State
q_0	0	0	R	q_0
q_0	#	0	R	q_1
q_1	0	0	R	q_1
q_1	B	B	L	q_2
q_2	0	B	N	q_3 (halt)

State Diagram

Execution Traces

Example 1: $2 + 3 \rightarrow 00\#000$

Initial Tape: 00#000

Steps:

1. q_0 , read 0 $\rightarrow$ move R
2. q_0 , read 0 $\rightarrow$ move R
3. q_0 , read # $\rightarrow$ write 0, move R $\rightarrow q_1$
4. q_1 , read 0 $\rightarrow$ move R
5. q_1 , read 0 $\rightarrow$ move R
6. q_1 , read 0 $\rightarrow$ move R
7. q_1 , read B $\rightarrow$ move L $\rightarrow q_2$
8. q_2 , read 0 $\rightarrow$ write B, halt $\rightarrow q_3$

Final Tape: 000000B $\Rightarrow$ represents 5 in unary



Figure 1: State Diagram of the Turing Machine for Unary Addition

Example 2: $5 + 1 \rightarrow 00000\#0$

Initial Tape: 00000#0

Steps:

1. q_0 , read 0 $\rightarrow$ move R ($\times 5$)
2. q_0 , read # $\rightarrow$ write 0, move R $\rightarrow q_1$
3. q_1 , read 0 $\rightarrow$ move R
4. q_1 , read B $\rightarrow$ move L $\rightarrow q_2$
5. q_2 , read 0 $\rightarrow$ write B, halt $\rightarrow q_3$

Final Tape: 000000B $\Rightarrow$ represents 6 in unary

Example 3: $5 + 6 \rightarrow 00000\#000000$

Initial Tape: 00000#000000

Steps:

1. q_0 , read 0 $\rightarrow$ move R ($\times 5$)
2. q_0 , read # $\rightarrow$ write 0, move R $\rightarrow q_1$
3. q_1 , read 0 $\rightarrow$ move R ($\times 6$)
4. q_1 , read B $\rightarrow$ move L $\rightarrow q_2$
5. q_2 , read 0 $\rightarrow$ write B, halt $\rightarrow q_3$

Final Tape: 00000000000B $\Rightarrow$ represents 11 in unary